Assignment Module-2

Course: B.Tech (CSE All) **Year/Semester:** II/IV **Session:** 2025-2026
Subject Name & Code: Operating Systems (BCSC0004)
Submission Date:29/11/2025

Q1: Consider the following page reference string. 1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,2 How many page faults would occur for the following replacement algorithm, assuming 3, 4 and 5 frames respectively?
a. Optimal page replacement. b. LRU page replacement. c. FIFO page replacement

Q2: Free memory holes of sizes 15K, 10K, 5K, 25K, 30K, 40K are available. The processes of size 12K, 2K, 25K, 20K is to be allocated. How processes are placed in first fit, best fit, worst fit. Calculate internal as well as external fragmentation.

Q3: Calculate page faults for (LRU, FIFO, OPT) for following sequences where page frame is three.
0,1,2,1,4,2,3,7,2,1,3,5,1,2,5.

Q4: System snapshot:

	Max	Allocation	Available
	A B C	A B C	A B C
P_0	0 0 1	0 0 1	
P_1	1 7 5	1 0 0	
P_2	2 3 5	1 3 5	
P_3	0 6 5	0 6 3	
Total		2 9 9	1 5 2

a) How many resources are there of type (A, B, C)?
b) What are the contents of the Need matrix?
c) Is the system in a safe state? Why?
d) If a request from process P1 arrives for additional resources of (0, 5, 2), can the Banker's algorithm grants the request immediately?
e) What would be the new system state after the allocation?

Q5: Consider a system consisting of four resources of the same type that are shared by three processes, each of which needs at most two resources. Is this system deadlock-free? Why or why not?

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Q6: Consider the following snapshot-

	Allocated				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Answer the following questions using banker's algorithm:
a) What are contents of matrix end?
b) Is the system in safe state?
c) If request for process p1 arrives for (0, 4, 2, 0). Can the request be granted immediately?

Q7. Given 3 processes A, B and C, three resources x,y and z and following events, i) A requests x ii) A requests y iii) B requests y iv) B requests z v) C requests z vi) C requests x vii) C requests y Assume that requested resources should always be allocated to the request process if it is available. Draw the resource allocation graph for the sequences. And also mention whether it is a deadlock? If it is, how to recover the deadlock.

Q8. A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for 4 processes are shown below:

Process	Maximum need	Current allocation
P1	9	3
P2	6	1
P3	5	3
P4	10	0

Describe the current status of the system as safe or unsafe.

Q9. Describe the SSTF disk scheduling algorithm using the following data. The dist head is initially at position-cylinder 53.the cylinder sequence of requests is 98, 183, 37, 122, 14, 124, 65. 67. Find the total head movement.

Q10. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive currently services a request at cylinder 143, and the previous request was at cylinder 125. the queue of pending request in FIFO order is 86,1470,913,1774,948,1509,1022,1750,130 Starting from the current position, what is the total distance(in cylinders) that the disk arm moves to satisfy all pending requests, for each of the following algorithms i)FCFS ii) SSFT iii) SCAN iv) LOOK v) C-SCAN.

Q11. The queue of requests in FIFO is 86, 147, 91, 177, 94,150,102,175,130 what is the total head movement needed to satisfy the requests for the following Scheduling algorithms FCFS, SJF, SCAN, LOOK, C-SCAN.

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Q12. Suppose that the head of moving head disk with 200 tracks numbered 0 to 199 is currently serving the request at track 143 and has just finished a request at track 125. If the queue request is kept in FIFO order, 86, 147, 91, 177, 94, 150, 102, 175, 130. What is the total head movement to satisfy these requests for i) FCFS II) SSTF disk scheduling algorithm

Q13: A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is _____.

Q14: Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

Q15: Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is _____ megabytes.

Q16: A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

Q17: Assume that there are 3-page frames which are initially empty. If the page reference string is 1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6, the number of page faults using the optimal replacement policy is_____.

Q18: Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is?

Q19: Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 On a demand paged virtual memory system running on a computer system that main memory size of 3 pages frames which are initially empty. Find the number of page faults under the corresponding page replacements policy.: LRU, FIFO and OPTIMAL.

Q20 A system uses FIFO policy for page replacement. It has 4 page frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?

Q21: A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1 If optimal page replacement policy is used, how many page faults occur for the above reference string?

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Q22: In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 16 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the following is the maximum number of bits that can be used for storing protection and other information in each page table entry?

Q23: Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order.

Perform the allocation of processes using-

1. First Fit Algorithm
2. Best Fit Algorithm
3. Worst Fit Algorithm

Q24. Consider a memory management system that uses a page size of 2 KB2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0,1,20,1,2, and 33 are stored in the page frames 1,3,21,3,2, and 00, respectively. The physical address (in decimal format) corresponding to the virtual address 25002500 (in decimal format) is ____

Q25. Consider the following two-dimensional array D in the C programming language, which is stored in row-major order:

```
int D[128][128];
```

Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet:

```
for (int i = 0; i < 128; i++)
    for (int j = 0; j < 128; j++)
        D[j][i] *= 10;
```

The number of page faults generated during the execution of this code snippet is _____.

Q26. Consider the following page reference string.

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

What are the minimum number of frames required to get a single page fault for the above sequence assuming LRU replacement strategy?

Q27. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each

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page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____

Q28. Determine the number of page faults when references to pages occur in the order 1,2,4,5,2,1,2,4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 brought earlier than page 2. (assume LRU i.e., Least-Recently-Used algorithm is applied).

Q29. Given reference to the following pages by a program 0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,3,8,3
How many page faults will occur if the program has three page frames available to it and uses an optimal replacement?

Q30. Consider a computer system with ten physical page frames. The system is provided with an accesses sequence (a1,a2,...,a20,a1,a2,...,a20), where each ai is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____.

Q31.A disk uses a block size of 512 bytes.A file of size 6,100 bytes is stored on the disk. *Find:*

- Number of disk blocks required
- Internal fragmentation in the last block.

Q32.A system uses a 24-bit virtual address space and a page size of 4 KB. Find the number of entries in the page table.

Q33.P={P1,P2,P3,P4} consists of all active processes in an operating system. R={R1,R2,R3,R4}R={R1,R2,R3,R4} consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: R1→P1,R2→P2,R3→P3,R4→P4Assignment edges: R1→P1,R2→P2 ,R3→P3,R4→P4 (the assignment edge R1→P1R1→P1 means resource R1R1 is assigned to process P1P1, and so on for others).
Claim edges: P1→R2,P2→R3,P3→R1,P2→R4,P4→R2Claim edges: P1→R2,P2→R3,P3→R1 ,P2→R4, P4→R2 (the claim edge P1→R2P1→R2 means process P1P1 is waiting for resource R2R2, and so on for others)

Q34. Consider the following state of a system with four processes, P1, P2, P3, and P4, and five types of resources, RS1, RS2, RS3, RS4, and RS5:

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C =

0	1	1	1	2
0	1	0	1	0
0	0	0	0	1
2	1	0	0	0

R =

1	1	0	2	1
0	1	0	2	1
0	2	0	3	1
0	2	1	1	0

E = (24144)

A = (01021)

Using the deadlock detection algorithm, show that there is a deadlock in the system. Identify the processes that are deadlocked.

35. Consider the following resource-allocation policy. Requests for and releases of resources are allowed at any time. If a request for resources cannot be satisfied because the resources are not available, then we check any threads that are blocked waiting for resources. If a blocked thread has the desired resources, then these resources are taken away from it and are given to the requesting thread. The vector of resources for which the blocked thread is waiting is increased to include the resources that were taken away.

For example, a system has three resource types, and the vector Available is initialized to (4,2,2). If thread T0 asks for (2,2,1), it gets them. If T1 asks for (1,0,1), it gets them. Then, if T0 asks for (0,0,1), it is blocked (resource not available). If T2 now asks for (2,0,0), it gets the available one (1,0,0), as well as one that was allocated to T0 (since T0 is blocked). T0's Allocation vector goes down to (1,2,1), and its Need vector goes up to (1,0,1).

a. Can deadlock occur? If you answer "yes," give an example. If you answer "no," specify which necessary condition cannot occur.

b. Can indefinite blocking occur? Explain your answer.

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Operating System Assignment

Q1- String 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2

Sol?-

1. FCFS with 3 frames

1	2	3	4	5	3	4	1	6	7	8	7	8	9	7	8	9	5	4	5	4	2
1	1	1	4	4			4	6	6	6		9				9	9			2	
	2	2	2	5			5	5	7	7		7				5	5			5	
		3	3	3			1	1	1	8		8				8	4			4	
*	*						*	*			*		*	*	*			*	*		

Page Hit = 8 9

2. LRU with 3 frames

1	2	3	4	5	3	4	1	6	7	8	7	8	9	7	8	9	5	4	5	4	2
1	1	1	4	4			4	4	7	7		7				5	5			5	
	2	2	2	5			1	1	1	8		8				8	4			4	
		3	3	3			3	6	6	6		9				9	9			2	
*	*						*	*			*		*	*	*			*	*		

Page Hit = 9

3. Optimal with 3 frames

1	2	3	4	5	3	4	1	6	7	8	7	8	9	7	8	9	5	4	5	4	2
1	1	1	1	5			5	5	7	7			7				5	5			5
	2	2	4	4			1	6	6	8			8				8	4			4
		3	3	3			3	3	3	3			9				9	9			2
*	*										*	*		*	*	*			*	*	

Page Hit = 9

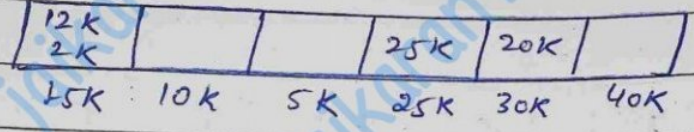
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Q2. Holes :- 15K, 10K, 5K, 25K, 30K, 40K

Processes :- 12K, 2K, 25K, 20K (P_1-P_4)

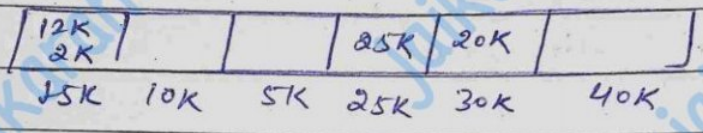
Sol? - i. First fit



E.f = 125-59 = 66K

I.f. = 3+1+10 = 14K

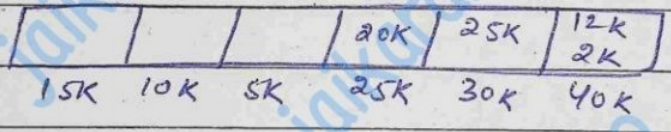
ii. Best fit



I.f. = 3+1+10 = 14K

E.f. = 125-59 = 66K

iii. Worst fit



E.f = 125-59 = 66K

I.f. = 20+20+5+5 = 84

Q4. a) Total resources = Allocated + Available

A = 2+1 = 3, B = 9+5 = 14, C = 9+2 = 11

Total resources = (3, 14, 11)

(b) Need = Max - Allocation.

P_0 = (0,0,0), P_1 = (0, 7, 5), P_2 = (1,0,0)

P_3 = (0,0,2)

Need ≤ work (available)

safe sequence → $P_0 - P_2 - P_1 - P_3$.

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(c) P_1 additional request of (0,5,2)

check, request ≤ need → (0,5,2) ≤ (0,7,2) = ok

request ≤ available → (0,5,2) ≤ (1,5,2) = ok

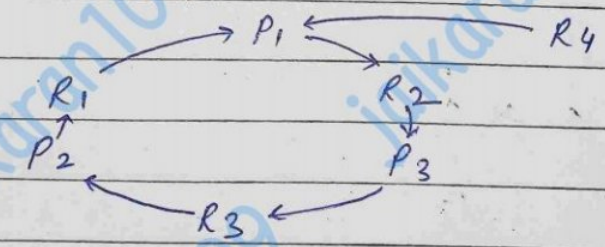
	Max			allocation			available	Need		
	A	B	C	A	B	C	A B C			
P_0	0	0	1	0	0	1	~~1 5 2~~	0	0	0
P_1	1	7	5	1	5	2	~~0 5 2~~	0	2	3
P_2	2	3	5	1	3	5		1	0	0
P_3	0	6	5	0	6	3		0	0	2
							1,0,0			

execution	available		
	A	B	C
P_0	1	0	1
P_1	2	3	6
P_2	3	8	8
P_3	3	14	11

Banker's algorithm will grant the permission -

(e) New state : $P_0 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$

Q5-



Yes, the system is deadlock free.

Because, firstly P_1 will be executed it has 2 resources, then once it freed the 2 resources P_3 & then P_2 will get resources & will be executed respectively.

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Q6. (a) need

	A	B	C	D
P_0	0	0	0	0
P_1	0	7	5	0
P_2	1	0	0	2
P_3	0	0	2	0
P_4	0	6	4	2

available = 1 5 2 0

(b) P_0 : (0, 0, 0, 0) ≤ (1, 5, 2, 0)

It will execute first.

A.R. = (1, 5, 3, 2)

P_2 : (1, 0, 0, 2) ≤ (1, 5, 3, 2)

new A.R. = (2, 8, 8, 6)

P_3 : (0, 0, 2, 0) ≤ (2, 8, 8, 6)

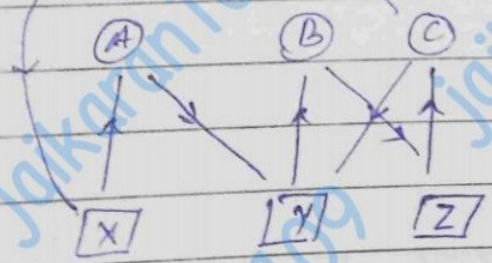
new A.R. = (2, 14, 11, 8)

P_4 : (0, 7, 5, 0) ≤ (2, 14, 11, 8)

new A.R. = (3, 14, 12, 12)

$P_0 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_1$

Q7 - Sol?



Yes, there is a deadlock,

A, B, C are holding X, Y, Z simultaneously & will infinitely wait for each other resources.

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Q8.

Process	max. need	allocation	Need
P_1	9	3	6
P_2	6	1	5
P_3	5	3	2
P_4	10	0	10
		7	

total resources = 9

Available = 9 − 7 = 2

P_3 : $2 \leq 2$

A.R = 2 + 3 = 5

P_2 : $5 \leq 5$

A.R. = 5 + 1 = 6

P_1 : $6 \leq 6$

A.R. = 6 + 3 = 9

P_4 will not executed as it do not satisfy the cond? need ≤ available

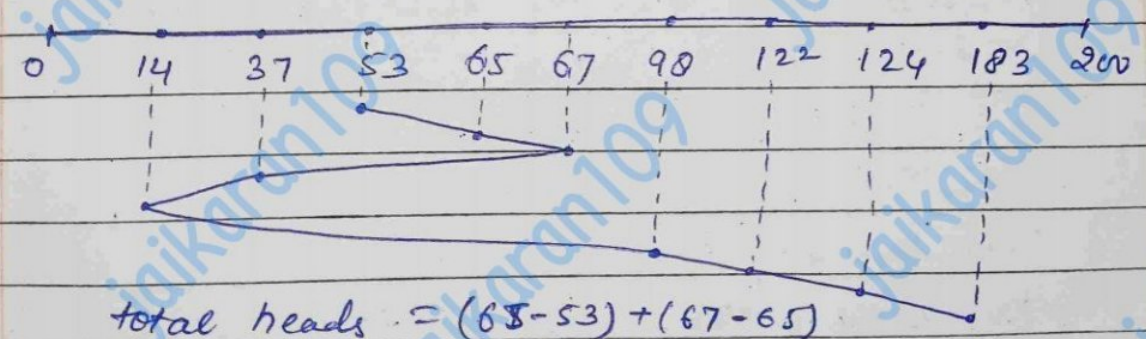
$9 \leq 10$ ✗

System is not in safe state.

Q9. SSTF.

requests : 98, 183, 37, 122, 19, 124, 65, 67

Posn : 53.



total heads = (65 − 53) + (67 − 65)
+ (67 − 37) + (37 − 14) + (183 − 14)
= 236

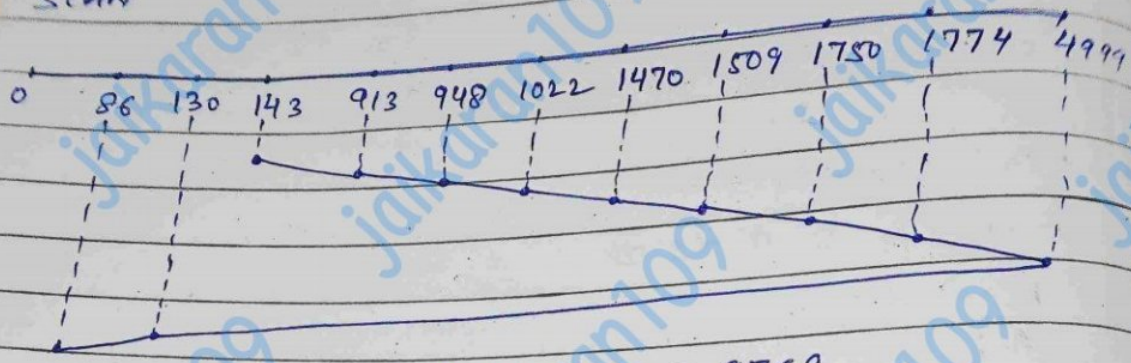
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Q10- requests: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

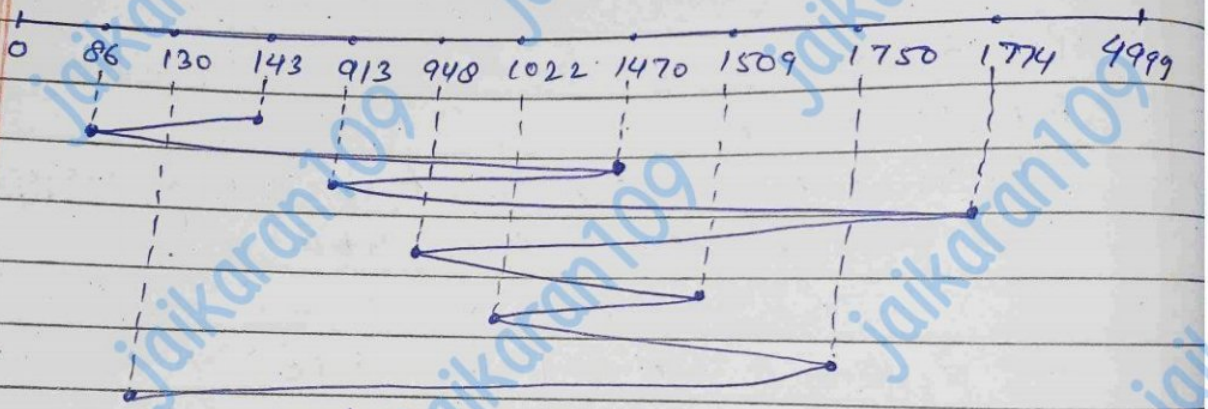
Pos": 143.

Sol? (i) SCAN



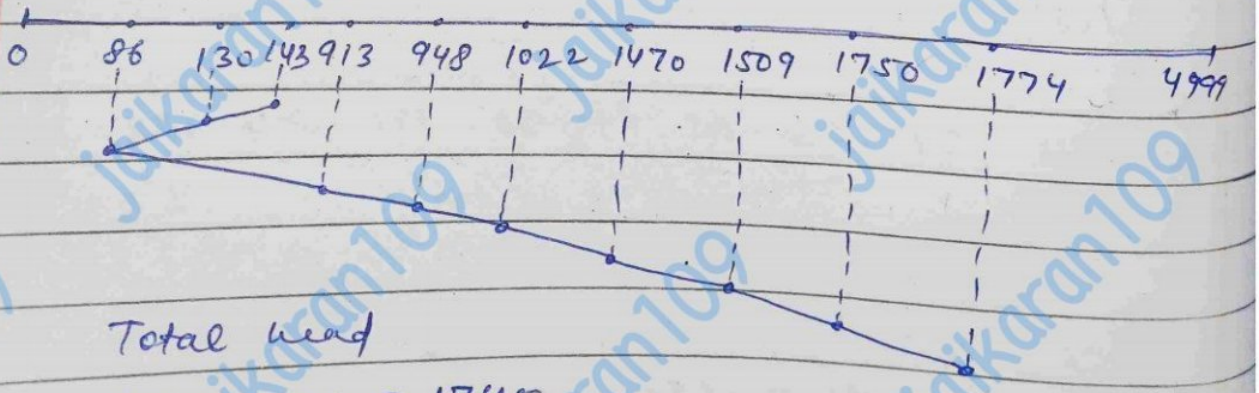
total head movement = 9769

(ii) FCFS.



head movement = 7081

(iii) SSTF



Total head movement = 1745

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Q11 requests: 86, 147, 91, 177, 94, 150, 102, 175, 130

pos°: 100

FCFS

Total head movement = 522

SSTF

Total head movement = 109

SCAN

Total head movement = 212

LOOK

Total head movement = 160

C-SCAN

Total head movement = 392

Q12- P = 3 (programs), n = 3 (instances)

Soln- No. of tape units that system must have such that deadlocks never arise is:

$$r \leq p(n-1)+1$$
$$\leq 3(3-1)+1$$
$$\boxed{r \leq 7}$$

Q14- LA = 64 bits, PA = 48 bits

Soln Page size = 8KB = $2^3 \times 2^{10} = 2^{13}$ B.

word size = 4B = 2^2 B

No. of distinct address = $\frac{\text{Page size}}{\text{word size}} = \frac{2^{13}}{2^2} = 2^{11}$ words

No. of valid entries in TLB = 128 = 2^7

No. of pages in TLB = 128

Total distinct virtual address = $2^7 \times 2^{11} = 2^{18}$

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Q15. virtual address = 40 bit

Sol? Page size = 16 KB = $2^4 \times 2^{10} = 2^{14}$ B

Page Table entry size = 48 bits = 6 bytes

No. of pages = $\frac{2^{40}}{2^{14}} = 2^{26}$

Page table size = No. of pages * PTE size

= $2^{26} \times 6$ bytes

1MB = 2^{20} bytes

= $\frac{2^{26} \times 6}{2^{20}} = 384$ MB

Q16. Page size = 8KB = $2^3 \times 2^{10} = 2^{13}$ B

Sol? Phy. address = 32 bits

PTE fields:

valid bit = 1, dirty bit = 1, permission bits = 3,

translation bits = PFN bits

Page offset = $\log_2(2^{13}) = 13$ bits

PFN bits = 32 - 13 = 19.

PTE size = 1+1+3+19 = 24 bits = 3 Bytes

No. of PTE = $\frac{24 \times 2^{20}}{3} = 2^3 \times 2^{20} = 2^{23}$.

valid pages = 2^{23}

virtual address bit = 36 bits.

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Q17. R.S. : 1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6

Sol? frame = 3

1	2	3	4	2	1	5	3	2	4	6
1	1	1	1			5	3			6
	2	2	2			2	2			2
		3	4	*	*	4	4	*	*	4

Page fault = 7

Q18. TLB = 10 + 80 = 90 m/s

Sol? TLB miss time = 10 + 80 + 80 = 170 ms

effective memory Access Time

EMAT = (% Time hit) + (1 − %) · Time miss

= (0.6 × 90) + (0.4 × 170)

= 122 ms

Q19. R.S. : 1, 2, 3, 2, 4, 1, 3, 2, 4, 1

Sol? frame = 3

(i) LRU

1	2	3	2	4	1	3	2	4	1
1	1	1	*	4	4	*	2	2	2
	2	2		2	1		1	4	4
		3		3	3		3	3	1

Page fault = 9

(ii) FIFO

1	2	3	2	4	1	3	2	4	1
1	1	1	*	4	4	*	4	*	*
	2	2		2	1		1		
		3		3	3		2		

Page fault = 9

(iii) optimal

1	2	3	2	4	1	3	2	4	1
1	1	1	*	1	*	*	1	*	*
	2	2		4			4		
		3		3			2		

Page fault = 5

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Q21. R.S : 1, 2, 1, 3, 7, 4, 5, 6, 3, 1

frame = 3

1	2	1	3	7	4	5	6	3	1
1	1	*	1	1	1	1	1	*	*
	2	2	2	7	4	5	6		
			3	3	3	3	3		

Page fault = 7

Q22. Virtual address = 32 bits

Phy. address = 30 bits

Page size = 16 KB = $2^4 \times 2^{10} = 2^{14}$ B

PTr size = 32 bits

memory is byte - addressable.

offset = $\log_2(2^{14}) = 14$ bits

PFN bits = 30 - 14 = 16 bits

Remaining bits in PTr for protection = 16 bits.

Q32. Virtual address space = 24 bits

Page size = 4 KB = $2^2 \times 2^{10} = 2^{12}$ B

LAS = 2^{24}

No. of entries in Page Table = $\frac{LAS}{PS}$

$= \frac{2^{24}}{2^{12}} = 2^{12}$

$= 4096$

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